

Kapil Sharma

☎ +91 7300761811 ✉ kapil.sharma.dev04@gmail.com 🔗 LinkedIn 🐙 Github

SUMMARY

Software Engineer with **4 years and 8 months** of experience building scalable microservices and event-driven systems in production environments. Core expertise in **Python (FastAPI, Django)**, event-driven architectures on **AWS**, and **PostgreSQL/NoSQL** data modeling. Experienced in production-grade **GenAI/LLM** integrations for workflow automation and decision support. Focused on system design and performance optimization.

TECHNICAL SKILLS

Languages: Python (Advanced), SQL, TypeScript

Backend & Architecture: FastAPI, Django Rest Framework (DRF), Celery, Microservices, Event-Driven Architecture, REST APIs, WebSocket

Cloud & DevOps: AWS (Lambda, ECS Fargate, SQS, EventBridge, CloudWatch, EC2, SES), Docker, Terraform (IaC), CI/CD (GitHub Actions)

Databases & Caching: PostgreSQL, DynamoDB, MongoDB, Redis, MySQL

AI & LLM Engineering: LangChain, LangGraph, OpenAI API, RAG Pipelines, Vector Embeddings, Ollama, LiteLLM, RAGAS, MCP

Protocols & Messaging: REST, WebSocket, MQTT, WebRTC, RabbitMQ, SQS

WORK EXPERIENCE

Tiger Analytics

Senior Application Engineer

Oct 2024 – Present
Bangalore, India (Hybrid)

- Designed and owned an event-driven automated email system using **AWS EventBridge** and **Amazon SES**, triggered every 15 minutes to process incremental claim updates and reliably deliver **approx 100k** emails per day.
- Designed, developed, and maintained **FastAPI-based microservices** supporting high-throughput claim processing in a production environment.
- Applied Test-Driven Development (TDD) practices by writing unit and integration tests, achieving 90%+ code coverage and significantly reducing regression defects.
- Contributed to the design and optimization of **PostgreSQL** schemas and **DynamoDB** data models to efficiently manage rule metadata, claim evidence, and audit logs.
- Built and engineered event-driven backend workflows using AWS Lambda and SQS, and integrated cloud-native services on AWS including **ECS Fargate, CloudWatch, and EC2** to improve scalability, observability, and operational stability.
- Integrated **LLM-based decision-support** components using structured prompts and retrieval-augmented context to assist complex claim validation and decisioning workflows within the auto-adjudication pipeline.
- Reviewed code contributions from team members, contributed to **CI/CD pipelines and infrastructure-as-code (IaC)**, and collaborated with cross-functional teams to define API contracts, test plans, and delivery milestones.

VVDN Technologies

Software Engineer

Sept 2021 – Oct 2024
Gurugram, India

- Designed and maintained robust **REST APIs using Django Rest Framework (DRF)**, improving throughput by 15–17% by optimizing SQL queries and refactoring legacy backend code.
- Developed backend features for Netgear's cloud-native Insight platform, using Python, PostgreSQL, and MongoDB to manage and orchestrate thousands of edge devices.
- Built real-time IoT integrations for NXP Home Assistant using **MQTT**, and implemented **WebRTC** signaling for low-latency video streaming dashboards.
- Collaborated with global engineering teams at NXP to identify, debug, and resolve production issues, improving system reliability and service availability.
- Played a key role in the development of an internal HR chatbot using **OpenAI APIs and RAG** to answer HR and company policy-related queries from internal documentation, while mentoring junior engineers and contributing to code reviews.

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University, Lucknow
B.Tech in Computer Science & Engineering

2017 – 2021
Percentage: 83.6